

A Virtual Platform for Overground Manual Wheelchair Navigation: a Controlled Ecological Approach for Exploring Wheeled Mobility.

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Figure 1: Large Overground Virtual Suite (LOGVs) example of manual wheelchair use in a virtual park

ABSTRACT

The overall aim of this research was to establish proof of concept for using the locally developed Large overground virtual suite (LOGVs) to explore manual wheelchair (MWC) mobility. The primary objective was to evaluate acceptability (system usability, presence, and confront) of LOGVs. The secondary objective was to explore visuolocomotor strategies used by able-bodied adults to circumnavigate a virtual pedestrian during MWC use versus bipedal walking. Using a controlled laboratory design, 15 participants completed multiple trials in LOGVs interacting with static and dynamic virtual pedestrians. LOGVs was acceptable, with participants reporting a high level of presence and perceived usability and a low level of cybersickness. Circumnavigation strategies were similar between bipedal walking and MWC use for clearance and deviation, as seen in previous research. Small differences observed during circumnavigation between MWC use and walking, such as looking less at the environment and more at the floor when using a MWC, may be attributed to the fact that participants were novice in MWC use. LOGVs seems to be an ecological and acceptable way to explore MWC mobility that may have future implications for evaluation and training.

Index terms: Manual wheelchair, visuolocomotor coordination, virtual reality.

1 INTRODUCTION

The circumnavigation of static obstacles has been studied during manual wheelchair (MWC) and powered wheelchair (propulsion and direction by upper limbs) movements in real-world environments [6,7]. Although locomotor adjustments in walking (propulsion and direction control by lower limbs) have been studied in virtual environments, few studies have explored MWC mobility in virtual reality (VR) [5,2,1]. Recently a Large Over Ground Virtual Suite (LOGVs) was created to explore multi-modal mobility over large spaces for both research and clinical applications [11]. To our knowledge, no study has focused on pedestrian circumnavigation in a MWC in a VR environment representative of a large real-world setting, such as a park. Moreover, locomotion directed by the upper limbs (as in MWC propulsion), has not yet been studied in the context of obstacle avoidance in virtual environments with the person moving around large spaces. Establishing proof of concept for studying MWC movement in VR will open the doors for exploring factors related to the social and physical environments during MWC navigation to help to understand how VR may be used for evaluation and training of MWC skills. The overall aim of this research was to establish proof of concept for using VR to explore MWC mobility. The primary objectives were to evaluate acceptability (usability, immersion and comfort). Secondary objective was to explore visuolocomotor strategies used by able-bodied adults to circumnavigate a virtual pedestrian during MWC use versus bipedal walking in LOGVs.

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2 METHODS

Design & Participants: An exploratory controlled laboratory design was used. Fifteen able-bodied adults were recruited. Inclusion criteria were: be between 18 and 50 years of age; have good visual acuity (greater than zero on the EDTRS® visual acuity test)[4]; have no physical or mental impairments that affect walking; have not recently experienced VR-related motion sickness.

Equipment and Data Collection: A virtual environment representing a park was created using Unreal Engine and projected onto a Quest Pro VR headset (90Hz). The virtual environment represented a park that measured 10 by 12.5 m, corresponding to the actual laboratory size. Objects were placed in the virtual environment at the exact locations where real objects were located to avoid risks of contact with real-world objects. Virtual pedestrians (one male and one female) were modeled using MetaHuman software and added to the virtual park. The stand-alone head mounted display (HMD) allowed participants to move freely within the entire laboratory space (LOGvs shown in Figure 1) [11]. A bird's view from inside the virtual park is shown in Figure 2.

Kinematic data from the HMD was collected to measure the distance between the participant and the virtual pedestrian. Additionally, the integrated eye tracker in the HMD (90Hz) was used to collect eye movement data. A MWC with anti-tip devices was used. A VR controller associated with the HMD was mounted on the MWC to allow real-time display of the MWC in the virtual environment and to calculate the distance between the MWC and a virtual pedestrian.

A Vicon motion capture system (Vantage 5, 90 Hz) was used to collect kinematic data using 36 reflective markers placed on the participant (upper trunk, acromion, C7, T10, sternum, manubrium and hands), the HMD and the MWC.

Procedure: The study consisted of a single session lasting approximately two hours. After obtaining written consent, participants completed self-administered questionnaires to assess preferred handedness (Handedness Questionnaire) [12], footedness (Waterloo Footedness Questionnaire) [8], and eyedness (Lateral Preference Inventory for Eyedness) [13].

Participants were given a short period to familiarize themselves with the MWC and the virtual park environment in LOGvs. During this time, participants first made slow head movements while standing stationary and then engaged in unconstrained walking between pre-set start and end points in the park. Walking speed was measured in each pathway over 5 meters using a stopwatch. Participants completed the familiarization when their walking speed stabilized (± 0.1 m/s), and they felt ready to start. The virtual pedestrian's walking speed was adjusted to 75% of the participant's walking speed. The participant was asked to walk toward a sign 8 meters ahead and read the message displayed (different message each trial).

The data collection consisted of two blocks: one for bipedal walking and one for MWC mobility. Each block included 5 trials without a virtual pedestrian, 10 trials with a static virtual pedestrian in the middle of the path, and 30 trials where the virtual pedestrian was walking toward the participant. In half of the trials, the virtual pedestrian deviated 15 degrees to the left or right to ensure their trajectory remained initially unpredictable.

Trials were presented in a pseudo-random order to prevent participants from anticipating conditions. Half of the participants started with bipedal walking, and the other half began with the MWC. Participants were instructed to read the sign while maintaining a comfortable walking speed and to deviate from the path of motion as needed. If they contacted the virtual pedestrian, the pedestrian verbally warned them. To minimize possible

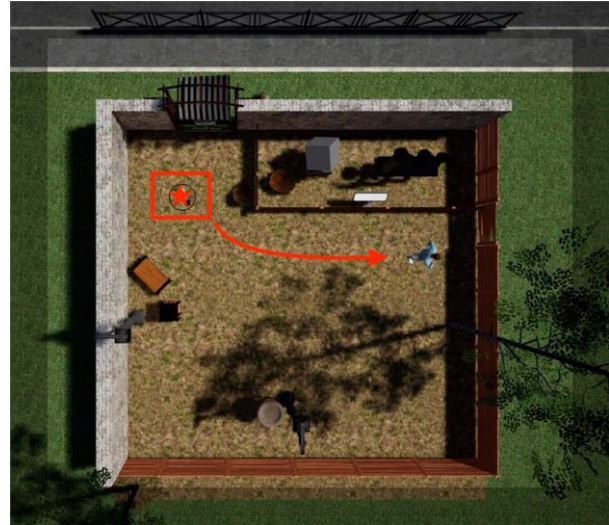


Figure 2: Birds eye view of the Large Overground Virtual Suite (LOGvs) and intended trajectory. The participant started in the square in both conditions and walked or wheeled toward the virtual pedestrian.

gender effects, participants chose the sex of the virtual pedestrian (male or female).

1° outcomes (acceptability): At the end of the session, participants completed three self-administered questionnaires to assess usability (System Usability Scale [9]), immersion (Presence Questionnaire [14]), and comfort (Cybersickness Questionnaire [10]).

2° outcomes (visuolocomotor strategies) :

- The minimal clearance between the participant and the virtual pedestrian, corresponding to the smallest distance between the participant's (or wheelchair's) extremities and the virtual pedestrian before turning.
- The side chosen for circumnavigation.
- The distance between the center of the virtual pedestrian and the center of mass of the participant at the time the deviation was initiated. This corresponded to the last moment when the medio-lateral velocity of the center of mass equaled zero, preceding a lateral displacement larger than two standard deviations compared to unconstrained walking.
- The percentage of time spent looking at various elements in the virtual environment before and after initiating the trajectory deviation (virtual pedestrian, target, environment, floor).

Data Analysis: Descriptive statistics were summarized. A non-parametric analysis (Kruskal-Wallis) was used to compare differences between walking and MWC use as data did not meet the parametric assumptions.

3 RESULTS

The sample comprised eight male and seven female participants, who had a mean (SD) age of 29.3 (6.3) years.

Level of presence and system usability were very high, with participants reporting greater than 5 out of 7 that the environment seemed natural, they were able to anticipate their movements in VR, and that they successfully interacted with the virtual environment and virtual pedestrians. The level of cybersickness was very low as participants reported less than 1 on all levels of cybersickness.

During circumnavigation, participants initiated path deviation at slightly greater (not statistically significant) distances from the pedestrian when walking than when using a MWC ($p=0.11$). There were no statistically significant differences in minimal clearance between the MWC and the virtual pedestrian during MWC use compared to walking, or the side chosen for circumnavigation.

When using the MWC, participants looked less at their surrounding environment ($p=0.08$) including the sign with the message they were instructed to read ($p=0.08$) and looked more at the floor compared to when they were walking ($p=0.007$).

4 DISCUSSION

Results of this study confirm proof of concept for using the LOGvs platform to explore MWC navigation, with a high level of perceived presence and usability and lower level of cyber sickness. Moreover, like Charrette et al. who used real world navigation in MWC and power wheelchairs, there were minimal differences between deviation and minimal distance of initial deviation between walking and MWC use, suggesting visuomotor strategies when using a wheelchair are similar to healthy walking [6,7]. Such similarities support the use of LOGvs for exploring MWC mobility (and possibly for exploring power wheelchair mobility in the future).

The slightly greater (but non-statistically significant) differences when walking may be explained by the fact that the current study recruited only able-bodied adults who had no previous MWC use experiences. Previous studies have shown that novice MWC users focus more on obstacles (i.e. the ground) when using MWC versus walking, suggesting there may be different information processing strategies for both locomotion modes (and especially with novice MWC users) [2].

Future studies will explore the acceptability of LOGvs with MWC users and clinicians. Then LOGvs may be used as a controlled and ecological way for exploring the influential factors of the physical and social environments during MWC navigation. LOGvs may also have future implications for evaluation and training for new MWC users.

5 CONCLUSION

Given the acceptability of LOGvs and the similarities between bipedal and MWC circumnavigation behaviours, proof of concept for LOGvs for exploring MWC mobility was achieved. VR in large overground spaces (LOGvs) may provide an ecological and acceptable way to explore MWC mobility, which may have future implications for evaluation and training.

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REFERENCES

[1] Alapakkam Govindarajan MA, Laplante-El Haili Y, Archambault PS. User validation of a portable manual wheelchair simulator. *Disabil Rehabil Assist Technol*. 2016;39(15):1-8.
 [2] Azimi S, Mostafavi MA, Best KL, Dommes A. Wheelchair users' navigational behavior: insights from eye movement data and environment legibility. 16th Conference on Spatial Information Theory (September 17-20, 2024, Québec City, Canada). *LIPICs COSIT 2024 proceedings*, 2024;315.

[3] Blouin M, Lalumiere M, Gagnon DH et al. Characterization of the immediate effect of a training session on a manual wheelchair simulator with haptic biofeedback: towards more effective propulsion. *IEEE Trans Neur Syst Rehab Eng* 2015;23(1):104-15.
 [4] Bühren, J. ETDRS Visual Acuity Chart. In: Schmidt-Erfurth, U., Kohnen, T. (eds) *Encyclopedia of Ophthalmology*. 2018: Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-540-69000-9_617
 [5] Chaar F, Archambault PS. Usability of a virtual reality manual wheelchair simulator. *Disability and Rehabilitation: Assistive Technology*. 2023; 18(8):1489-1499.
 [6] Charette C, Routhier F, McFadyen BJ. Visuo-locomotor control in persons with spinal cord injury in a manual or power wheelchair for direction change and obstacle circumvention. *Exp Brain Res*. 2017;235(9):2669-2678. doi:10.1007/s00221-017-4994-4
 [7] Charette C, Routhier F, McFadyen BJ. Visuo-locomotor coordination for direction changes in a manual wheelchair as compared to biped locomotion in healthy subjects. *Neurosci Lett*. 2015;588:83-87. doi:10.1016/j.neulet.2015.01.002
 [8] Elias LJ, Bryden MP, Bulman-Fleming MB. Footedness is a better predictor than is handedness of emotional lateralization. *Neuropsychologia*. 1998;36(1):37-43. doi:10.1016/S0028-3932(97)00107-3
 [9] Grier, R. A., Bangor, A., Kortum, P., & Peres, S. C. (2013). The System Usability Scale: Beyond Standard Usability Testing. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 57(1), 187-191. <https://doi.org/10.1177/1541931213571042>
 [10] Kourtesis, P.; Linnell, J.; Amir, R.; Argelaguet, F.; MacPherson, S.E. Cybersickness in Virtual Reality Questionnaire (CSQ-VR): A Validation and Comparison against SSQ and VRSQ. *Virtual Worlds* 2023, 2, 16-35. <https://doi.org/10.3390/virtualworlds2010002>
 [11] McFadyen BJ, Blanchette AK, Best KL, Caron-Roberge J, Boulo J, LogVS Team. Democratizing a Large Over Ground Virtual Suite (LogVS) for Rehabilitation Sciences and Practice Exploiting Human Interaction. *IEEE VR 2024*. March 16-21, 2024. Orlando, Florida, USA.
 [12] Oldfield, RC. The assessment and analysis of handedness: The Edinburgh inventory. *Neuropsychologia*. 1971;9(1):97-113. doi:10.1016/0028-3932(71)90067-4
 [13] Coren, S. "The lateral preference inventory for measurement of handedness, footedness, eyedness, and earedness: Norms for young adults." *Bulletin of the psychonomic society* 31 (1993): 1-3.
 [14] Witmer, B. G., & Singer, M. J. (1998). *Presence Questionnaire (PQ)* [Database record]. APA PsycTests. <https://doi.org/10.1037/t21619-000>